

hit 'Play animation'. All the vertical objects have the same animation. The base object keyframes can now be modified. Remember we had used the Instance option while creating an array. Let's see how to use this feature. Select the 3rd object in the bottom row. Observe that the keyframes are the same for the objects placed vertically above it. Modify the keyframe position of the base object. All the keyframes in the instanced objects above it are updated automatically. Group all the objects using the group command in the menu bar. Save this file as tech4.max.

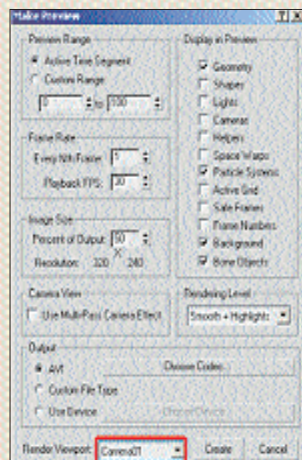
Using Instance while copying, allows you to modify the parameters of the master object. All instances created from the master object are updated automatically. This is useful when developing animation—say, with a school of fish, where you modify one fish to update all instanced objects.

Camera animation

Create a free camera in the Front viewport from the camera command panel. Move the camera along the Y-axis so as to view the grouped objects. Select Camera view in Perspective viewport or press [C] while there. Your animation is displayed as seen through the camera.

Move the camera till the objects just fit to the extremes of the view. This is your view in Frame 1. Now, turn on AutoKey and go to frame 25.

Move the camera in the



Quickly check your animations in Preview mode

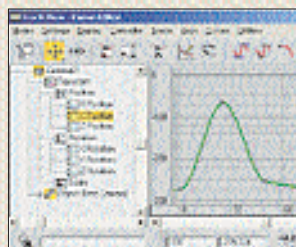
left view along the X-axis closer to the object. Move it along the same axis away from the object in frame 50. Use your own creative camera compositions. Repeat the same for frame 75. For frame 100, copy the keyframe from frame 1 to frame 100 by clicking and dragging the frame 1 marker, while holding down [Shift]. Your basic animation is ready. You can preview your animation by selecting the **Animation > Make Preview** command from the menu bar. Save the file as tech5.max

The Make Preview window has options that allow you to set the size and parameters of your preview. Ensure that you have chosen Camera01 from the Render Viewport drop-down option. You can also create multiple cameras in the same scene and animate them. You can name these cameras for your convenience. Set the camera name during preview in the Render Viewport dropdown.

Adding dynamism to camera animation

Regular keyframing gives you a very standard looking animation. You can add spice to your animation by using **Track View > Curve editor**.

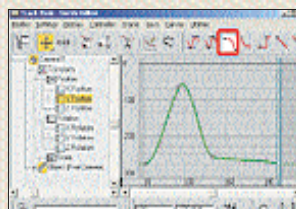
Select the camera and select the **Track View > Curve editor** from the Graph Editors menu to open the curve-editor window. Select the Transform Y position of the camera in the left panel of the editor window. A green curve defines the movement of the camera. It starts with an ease in, takes a peak and then ends with an ease out. This is the



Curves that can set things right

default attribute assigned to animated objects in 3dsmax. However, we can assign different controls to change the nature of animation required.

You can modify the nature of curve in the graph editor to change the tempo of the animation. Click on the second node to display the



Adjust the curve to change the pace of the animation

Bezier control handles. Drag the left node vertically down to change the nature of curve. Preview the animation using the camera viewport.

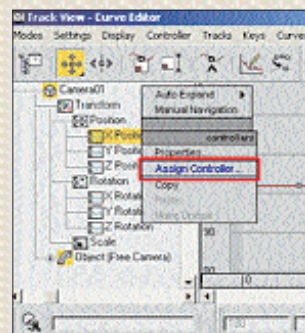
Similarly, select the fourth node and select the Set Tangent to Fast icon on the curve-editor top panel. Preview it to see the difference. Save the file as tech6.max.

Handheld camera effect

Assign controllers to the X or Z-axis to make the camera movements look wacky or like the effect of a jerk when using a handheld camera.

Assign a controller by right-clicking on the X position of camera01, transform the position in the curve-editor, and select 'Assign controller' to open the Assign Float controller box. Select 'Noise Float' and click OK.

An irregular wave pattern appears in the Noise Con-



Tweak the curve controls to get that jerky look

troller property window. Preview the animation. Observe that your camera has shifted to the left and the movement is a bit jerky. Adjust the Group01 object



Virus Alerts, Bug Fixes, Patches, Security

Virus Workshop

www.zdnetindia.com/digital